Rate floor stable across a 20× Δt sweep

exp044 · 2 June 2026

The trained networks this entry uses are produced once in the shared training hub, exp022 (Training), and reused here rather than retrained.

Abstract

The Δt audit asks whether the exp025 headline E rate is a physical (Hz) property of the trained network or an artefact of the integration timestep. The rate stays in a 9.5–14 Hz band across a 20× Δt sweep, accuracy holds at 88.5–90.7%, and the gamma cycle period in physical ms is invariant — but the rate dependence on Δt is non-monotonic, with Δt = 0.5 ms producing a *lower* mean rate than Δt = 0.25 ms.

Method

Parameter	Value
Integration timestep Δt	0.1 ms
Trial duration T	200 ms
MNIST samples (80/20 stratified split of 2000)	1600 train / 400 test (≈ 2.9% of the 70k-sample MNIST corpus)
Epochs	100

Train one PING from scratch per $\Delta t \in \{0.05, 0.1, 0.25, 0.5, 1.0\}$ ms × seed $\in \{42, 43, 44\} = 15$ cells. Total physical time $T = 200$ ms held constant (step count varies 4000 → 200). Batch size 64 throughout — smaller than exp025′s 256, but matched across the sweep so per-step compute and memory stay comparable, and the Δt = 0.05 cells (4000 timesteps × N_E × N_I) fit in a single A100. All other PING recipe parameters held to exp025.

After training, run inference on the test set; report mean E rate (Hz), accuracy, and a single-trial raster from seed 42 per Δt for visual cycle-period inspection.

Results

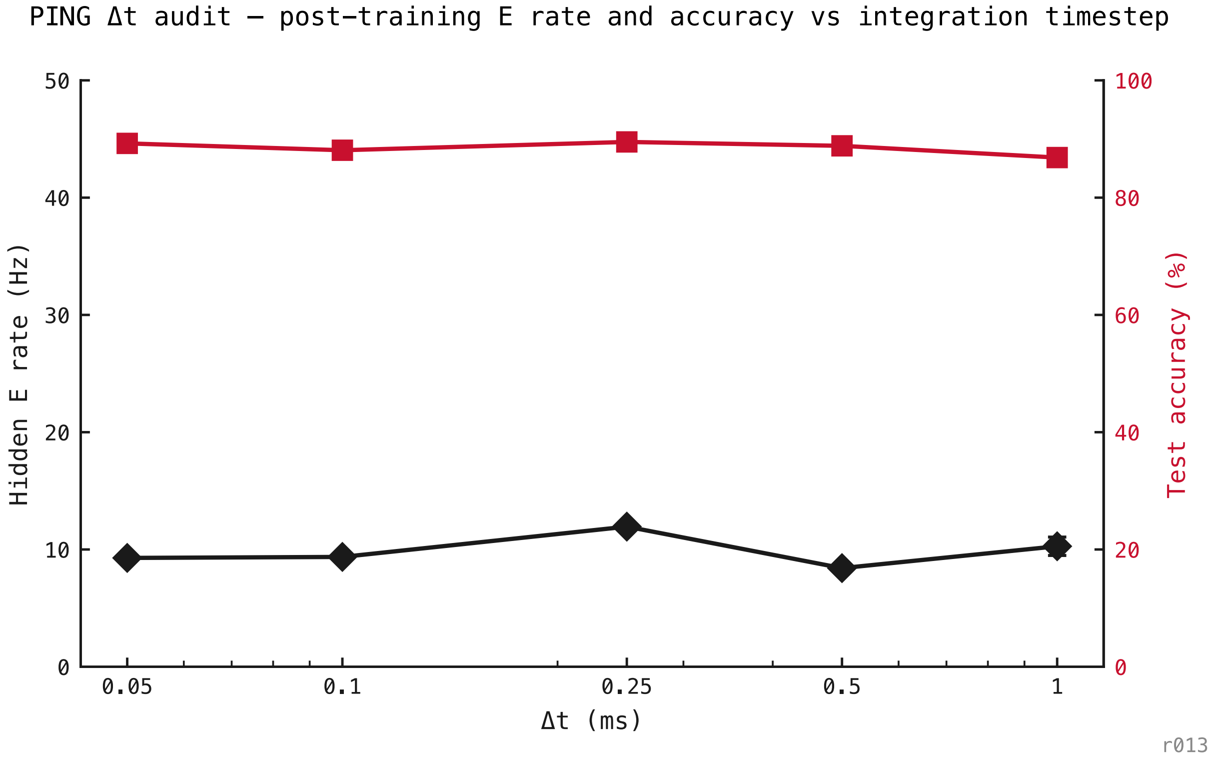

Figure 132: Hidden E rate (black) and test accuracy (red) as Δt varies 20$\times$ on a log scale. Error bars from three seeds.

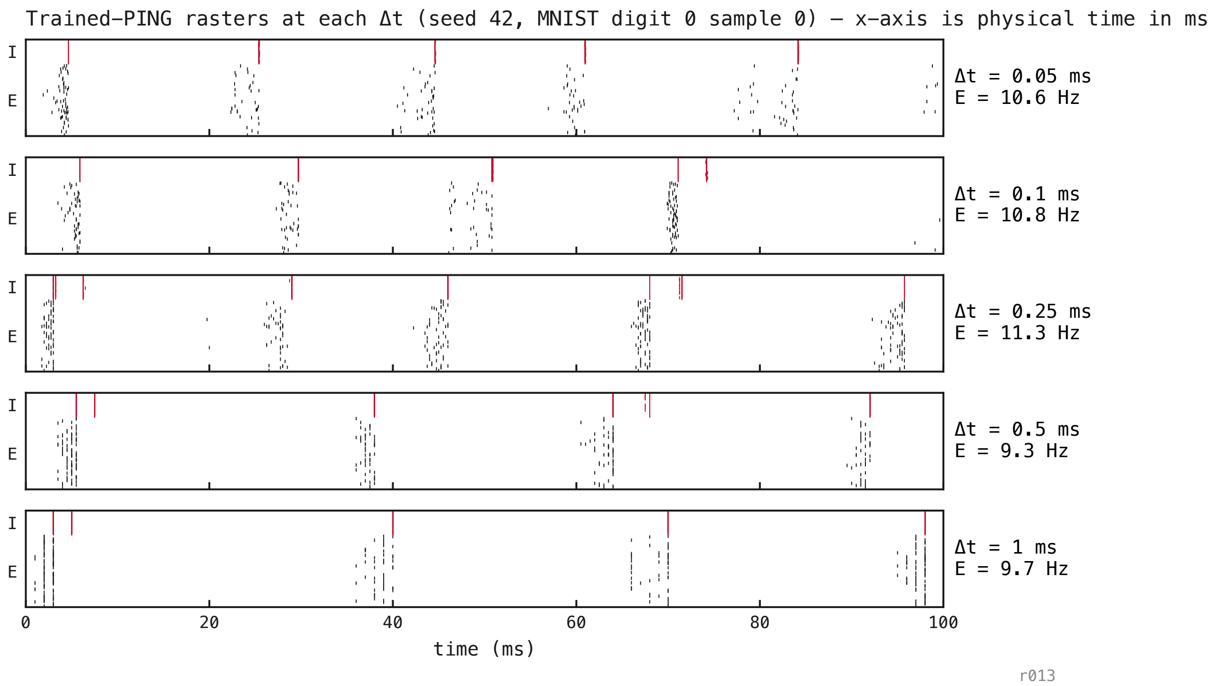

Figure 133: Single-trial rasters at each Δt, x-axis in *physical* ms (not steps). All five panels show E and I bursts at the same gamma cadence ($\approx$ 30 ms cycle). The cycle physics is Δt-invariant.

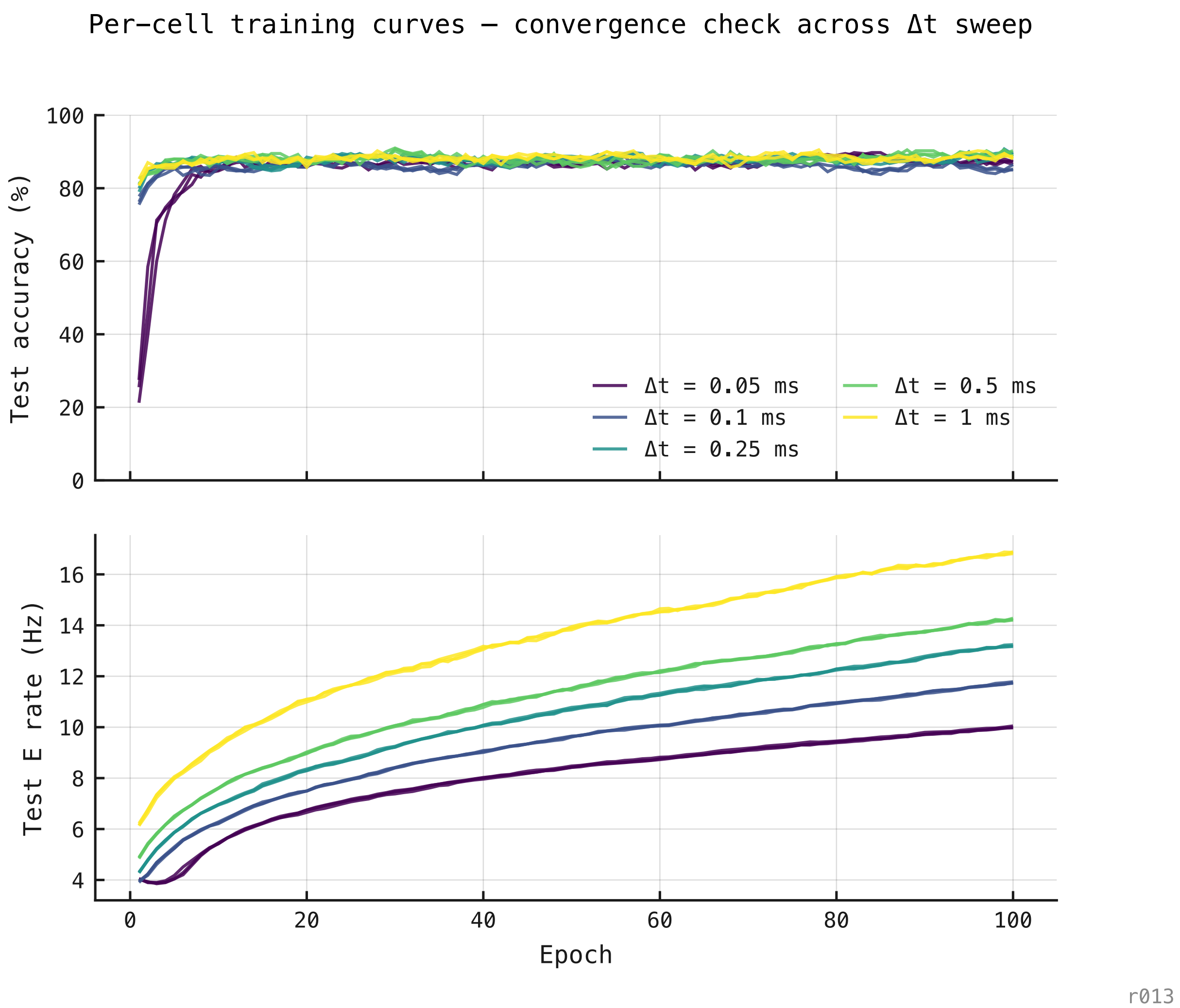

Figure 134: Top: test accuracy converges by epoch ≈ 10–15 across all Δt. Bottom: test E rate has *not* converged in 30 epochs — every curve is still rising. The Δt-dependent rate scaling is a snapshot of medium-tier training at epoch 30, not a fixed-point ceiling.